

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Momentum, Impulse & Collisions

*Momentum questions are bookkeeping: direction, before/after,
and impulse.*

MOMENTUM, IMPULSE & COLLISIONS · M1 · 09

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 MOM: Conserve Momentum

TRIGGER *collision, before and after*

02 IMP: Impulse Change

TRIGGER *impulse, change in momentum*

03 COLL: Collision Pair

TRIGGER *two particles collide*

04 FORCE: Impulse From Force

TRIGGER *constant force for time, force-time graph*

BANK EVIDENCE Local pages: 29 problem, 51 answer, 9 notes. Website primary entries: 18. Website linked entries: 22.

01 MOM: Conserve Momentum

TRIGGER *collision, before and after*

BECOMES Total momentum before equals total momentum after, in one chosen direction.

FIRST LINE TO WRITE

$$\sum mu = \sum mv$$

SIMPLEST STRATEGY

1. Mark positive direction.
2. Order before and after.
3. Multiply mass by velocity.
4. Equate totals.
5. Negative signs for opposite direction.
6. Tidy unknown velocity.

WORKED MODEL

$$2(5) + 3(-1) = 2v_1 + 3v_2.$$

HARD VARIANTS

1. Use velocities, not speeds, inside the equation.
2. External impulse must be absent for conservation.
3. Check if particles coalesce and share one final velocity.

— BOTTOM LINE

Momentum conservation is signed.

02 IMP: Impulse Change

TRIGGER *impulse, change in momentum*

BECOMES Impulse equals final momentum minus initial momentum.

FIRST LINE TO WRITE

$$I = m(v - u)$$

SIMPLEST STRATEGY

1. Identify direction.
2. Multiply final velocity.
3. Prepare initial momentum.
4. Use final minus initial.
5. Label N s.
6. State direction.
7. Evaluate.

WORKED MODEL

■ If $u = 4$ and $v = -2$, then $I = m(-2 - 4)$.

HARD VARIANTS

1. The impulse on the other body is equal and opposite.
2. Units are N s or kg m s^{-1} .
3. If a force-time graph is given, impulse is area.

— BOTTOM LINE

Impulse is a signed change.

COMMON TRAP Using $u - v$ instead of $v - u$.

03 COLL: Collision Pair

TRIGGER *two particles collide*

BECOMES Use momentum plus any given collision relation.

FIRST LINE TO WRITE

| momentum equation + second relation

SIMPLEST STRATEGY

1. Choose direction.
2. Organise before/after table.
3. Link with momentum.
4. Link with given condition.
5. Solve simultaneous equations.
6. Decide physical root.
7. End with speeds/directions.

WORKED MODEL

| If they move together, set $v_1 = v_2$ after impact.

HARD VARIANTS

1. For coalescence, both final velocities are equal.
2. If a rebound happens, one velocity changes sign.
3. Check that post-collision directions match the signs you report.

— BOTTOM LINE

One equation is rarely enough for two final velocities.

COMMON TRAP Forgetting the second condition.

04 FORCE: Impulse From Force

TRIGGER *constant force for time, force-time graph*

BECOMES Use impulse as force times time or area under force-time graph.

FIRST LINE TO WRITE

$$I = Ft \quad \text{or} \quad I = \text{area}$$

SIMPLEST STRATEGY

1. Find the time interval.
2. Obtain force or graph area.
3. Relate to $m(v - u)$.
4. Calculate unknown.
5. Express direction.

WORKED MODEL

■ A force 6 N for 0.4 s gives impulse 2.4 N s.

HARD VARIANTS

1. For triangular force graphs, use triangle area.
2. Direction of force controls sign of impulse.
3. Average force is impulse divided by time.

— BOTTOM LINE

Impulse is the area under a force-time graph.

COMMON TRAP Using acceleration without the time interval.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
collision, before and after	$\sum mu = \sum mv$
impulse, change in momentum	$I = m(v - u)$
two particles collide	momentum equation + second relation
constant force for time, force-time graph	$I = Ft$ or $I = \text{area}$

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
MOM: Conserve Momentum	Use velocities, not speeds, inside the equation. External impulse must be absent for conservation. Check if particles coalesce and share one final velocity.
IMP: Impulse Change	The impulse on the other body is equal and opposite. Units are N s or kg m s ⁻¹ . If a force-time graph is given, impulse is area.
COLL: Collision Pair	For coalescence, both final velocities are equal. If a rebound happens, one velocity changes sign. Check that post-collision directions match the signs you report.
FORCE: Impulse From Force	For triangular force graphs, use triangle area. Direction of force controls sign of impulse. Average force is impulse divided by time.